

Partially-Coupled Staircase LDPC Codes for High-Speed Inter-Satellite Communications

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Abstract—In this paper, we propose a rate-compatible partially-coupled staircase low density parity check (PS-LDPC) coding scheme for high speed inter-satellite communications. First, we introduce the encoding process and sliding window decoding (SWD) algorithm of PS-LDPC codes, and we investigate the error floor of component codes, which validate that the PS-LDPC codes with short block-length component code can maintain the reliability, and significantly reduce the decoding latency. Then, we analyze the density evolution (DE) of PS-LDPC codes based on the multi-edge type (MET)- LDPC framework under the Gaussian approximation, and derive its decoding thresholds of SWD. Further, we propose an optimized coupling pattern (OCP) encoding algorithm that achieves the optimal coupling patterns with the minimized threshold by introducing two-stage column permutations, and modify the message exchanges in SWD algorithm according to this encoding algorithm. Moreover, we design a new decoding algorithm, named cascaded SWD (C-SWD) algorithm, which reduces the error floor and enhances decoding performance by pre-decoding, reliability enhancement, and cascading belief propagation (BP) decoder or ordered likelihood decoder (OLD) due to the error floor. Simulation results demonstrate that our PS-LDPC coding scheme outperforms the existing rate-compatible spatially coupled LDPC (SC-LDPC) coding schemes in terms of bit error rate and complexity.

Index Terms—Inter-satellite communications, staircase LDPC codes, density evolution, sliding window decoding, rate-compatible SC-LDPC codes, error-rate performance.

I. INTRODUCTION

WITH the rapid deployment of large-scale low Earth orbit (LEO) satellite constellations, inter-satellite communications have become vital to enable the upcoming satellite-integrated Internet, aiming to reduce reliance on

ground stations and enable seamless global coverage [1], [2], [3], [4], [5]. Nevertheless, the realization of reliable high speed inter-satellite communications faces several challenges, including long free space path loss [6], [7], limited onboard transmission power [8], satellite motion, pointing errors [9], background light, and various cosmic rays [10], [11]. Therefore, rate-compatible forward error correction (FEC) schemes are essential, allowing adaptive coding to ensure both reliability and efficiency under dynamic link conditions [12], [13].

Staircase codes [14] are adopted in the ITU-T G.709.2/Y.1331.2 standard [15], and have been studied in recent studies for inter-satellite and deep space communication systems [16], [17]. Constructed with Bose-Chaudhuri-Hocquenghem (BCH) component codes [18], [19], [20] to form ladder structures, staircase codes support efficient parallel encoding and decoding, making them well-suited for high speed applications. In fiber-optical communication systems, staircase codes cascading with Hamming codes have been employed to achieve throughput designs of 400 Gbps [21] and even 21.77 Tbps [22]. However, existing schemes struggle to satisfy the stringent reliability demands of inter-satellite communications, and the limited flexibility of BCH code parameters [23] restricts dynamic adjustment of code rates and block-lengths.

Hence, other block codes such as low density parity check (LDPC) [24], [25], [26] and polar codes [27], [28] have been explored as potential component codes for staircase codes. Staircase LDPC codes have demonstrated performance gains of 0.5-1 dB over staircase BCH codes when using the belief propagation (BP)-based sliding window decoding (SWD) algorithm with bit-flipping [24]. Specifically, various rate-compatible spatially coupled LDPC (SC-LDPC) codes have been proposed [29], [30], [31], [32]. These codes offer encoding efficiencies comparable to their component codes and adapt SWD with tunable latency, which makes them attractive for high speed inter-satellite communications. The authors in [30] propose partially information-coupled LDPC (PIC-LDPC) codes and corresponding feed-forward feed-back (FF-FB) decoding algorithm, where the code rate can be adjusted by a simple coupled information bits. The authors in [31] propose systematic convolutional low density generator matrix (SysConv-LDGM) codes and corresponding iterative SWD (ISWD) algorithm, where the code rate can be adjusted by the encoding memory. The authors in [32] propose partial spatially coupled LDPC (PSC-LDPC) codes, where the code rate can be adjusted by modifying the proportion between block LDPC protographs and SC-LDPC protographs, and a large parity check matrix is constructed via the progressive edge-growth algorithm [33]. Therefore, an alternative coding scheme is need to balance performance and complexity.

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A preliminary study on rate-compatible staircase LDPC codes is presented in [25], but it lacks detailed theoretical analysis and decoding algorithm. On one hand, for spatially coupled codes, different coupling patterns can lead to distinct performance. The coupling pattern of PIC-LDPC codes [30] is optimized using the extrinsic information transfer (EXIT) chart [34]. The decoding threshold of sub-block rearranged staircase codes [19] under a fixed coupling pattern is analyzed via density evolution (DE) [35], [36], [37], [38], which is effectively optimized the threshold even with small length component codes, and the conditions for achieving better threshold are derived. However, there is still a lack of research on analyzing the threshold of partially-coupled staircase LDPC (PS-LDPC) codes and optimizing the coupling patterns. On the other hand, unlike schemes that protect encoded bits by multiple component codes [39], PS-LDPC codes contain uncoupled bits protected by only one component codeword, which may degrade overall performance. Recently, the authors in [40] propose the ordered likelihood decoder (OLD) for short rate-compatible codes, which re-encodes the most reliable basis (MRB) to correct errors based on an integer partitions-aided test error patterns (TEPs) generating criterion. This decoder is particularly applicable to PS-LDPC codes, as it takes advantage of the high reliability of coupled bits to correct the more error-prone uncoupled bits. Motivated by the need for both high performance and flexibility in inter-satellite communications, we further investigate PS-LDPC codes with improved algorithms in this paper.

In this paper, we propose a rate-compatible PS-LDPC codes and investigate coding algorithms to better support high-speed inter-satellite communications, aiming to meet the reliability requirements of bit error rate (BER) = 10^{-5} [41] or BER = 10^{-6} [11], [42] at lower signal to noise ratio (SNR). The main contributions of this work are summarized as follows.

- To meet the Gbps+ throughput requirement of inter-satellite communications under varying channel conditions, we propose the PS-LDPC codes, which support flexible code rates by adjusting the coupling depth. Then, we analyze the error floors of short block-length component codes under SWD algorithm, and approximate the block error rate (BLER) lower bound of component codes with ideal a priori information. This analysis validates the short block-length component codes can maintain the reliability and significantly reduce the decoding latency, which gives a practical guideline for subsequent optimization.
- Then, we characterize the PS-LDPC code ensemble using the multi-edge type (MET)-LDPC framework [37] with three edge-types, and define the node-perspective polynomials of variable nodes (VNs) and check nodes (CNs). Under the Gaussian approximation, we describe the DE process of SWD algorithm by tracking the mean and variance of messages, and provide the threshold result. Furthermore, by introducing the two-stage column permutations to adjust the edge-types connected to each VN, we design an optimized coupling pattern (OCP) encoding algorithm that achieves the optimal coupling patterns with the minimized threshold, and modify the message exchange of SWD algorithm according to this encoding algorithm.

- Moreover, we propose the cascaded SWD (C-SWD) algorithm to select BP or OLD due to the error floor, which improves decoding performance by pre-decoding, reliability enhancement, and cascaded decoders. Further, we analyze the complexity of our PS-LDPC coding scheme, and compare to the existing schemes by counting the number of binary additions, real additions, multiplications, comparisons, and table look-up operations. Simulation results demonstrate that our PS-LDPC coding scheme outperforms the related rate-compatible SC-LDPC coding schemes in terms of BER and complexity.

The remainder of this paper is organized as follows. Section II introduces the fundamentals of PS-LDPC codes, including the encoding process, SWD algorithm, and preliminary simulation results. In Section III, the PS-LDPC code ensemble is characterized, followed by DE approximate analysis and corresponding threshold results. Section IV of this paper describes the proposed algorithm optimization, including the OCP encoding algorithm and C-SWD algorithm, along with a comprehensive complexity analysis. Simulation results are presented in Section V, where we compare the BER and complexity of our proposed scheme with existing schemes, and separately evaluate the performance improvements from the OCP encoding algorithm and C-SWD algorithm. Finally, Section VI concludes the paper.

II. PS-LDPC CODES

In this section, we first introduce the encoding process of PS-LDPC codes. Next, we describe the dual message-passing iteration process for the corresponding SWD algorithm. Finally, we analyze the performance under short block-length component LDPC codes, which provides critical insights for optimizing decoding algorithm to lower the error floor.

A. Encoding Process

The proposed PS-LDPC codes are a class of spatially coupled product-like codes [14]. Each staircase block is constructed by jointly and in parallel encoding a portion of the previous block together with a new information block. The resulting parity check block, along with the information block, forms a complete new staircase block.

The encoding structure of PS-LDPC codes is shown in Fig. 1. First, we select a systematic LDPC code $\mathcal{C}(N, K)$ as the component code, with the codeword length of N , the message bits length of K , the parity check bits length of $M = N - K$, and the generator matrix $\mathbf{G}$. Let $\mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_i, \dots, \mathbf{B}_L, \mathbf{B}_{t1}, \mathbf{B}_{t2}, \dots, \mathbf{B}_{t\Upsilon}$ denote the matrices of size $m \times (N - m)$, where m is the coupling depth, $\mathbf{B}_i$ for $1 \leq i \leq L$ are the i -th message blocks, and $\mathbf{B}_{t1}, \mathbf{B}_{t2}, \dots, \mathbf{B}_{t\Upsilon}$ are Υ termination blocks. For each $\mathbf{B}_i = [\mathbf{B}_{i, \text{info}}, \mathbf{B}_{i, \text{parity}}] = [\mathbf{B}_{i, L}, \mathbf{B}_{i, R}]$, $\mathbf{B}_{i, \text{info}}$ is the submatrix of information bits with size $m \times (K - m)$, and $\mathbf{B}_{i, \text{parity}}$ is the submatrix of parity check bits with size $m \times M$.

Then, the encoding process of PS-LDPC codes starts from $\mathbf{B}_1$ and proceeds sequentially in a forward order. As shown in Fig. 1, the matrix $\mathbf{C}_0$ is initialized to an all-zero matrix known to the receiver, and does not send over the channel. Moreover, $\mathbf{B}_{i, \text{parity}} = [\mathbf{C}_{i-1}, \mathbf{B}_{i, \text{message}}] \cdot \mathbf{G}$ is obtained by encoding row by row in parallel, where $\mathbf{C}_{i-1} = \mathbf{B}_{i-1, R}^T$ of size $m \times m$

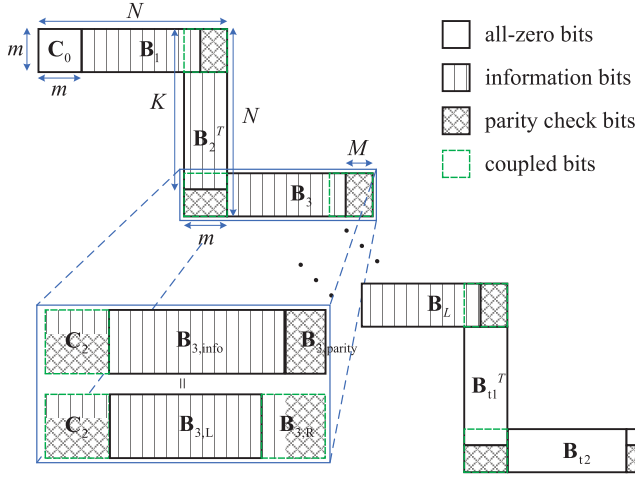


Fig. 1. The encoding structure of PS-LDPC codes with L message blocks and $\Upsilon = 2$ termination blocks.

represents the coupled bits matrix for $\mathbf{B}_i$, T represents the matrix transpose. The parameter m also determines the level of parallel processing, as all m rows of a block can be encoded or decoded simultaneously. The termination blocks are used to ensure reliable decoding of the terminal message blocks.

For each message block, the number of transmitted information bits is $(K - m) \times m$, and the number of transmitted parity check bits is $M \times m$. For termination blocks, no information bits are transmitted. For the first termination block, the number of transmitted parity check bits is $M \times m$. For the second and subsequent termination blocks, if $m < M$, the number of transmitted parity check bits is $m \times m$; otherwise, it remains $M \times m$. Thus, the actual code rate of PS-LDPC code with L message blocks and Υ termination blocks is

$$R = \frac{(K - m)mL}{(N - m)mL + mM + (\Upsilon - 1)M \cdot \min(m, M)}, \quad (1)$$

where $(K - m)mL$ is the total number of information bits transmitted, $(N - m)mL$ is the total number of transmitted bits in the L message blocks, mM is the number of transmitted bits in the first termination block, and $(\Upsilon - 1)M \cdot \min(m, M)$ is the total number of transmitted bits in the second and subsequent termination blocks.

For continuous streaming information sources [14], the code rate of PS-LDPC codes without termination, i.e., design rate, is

$$R_d = \lim_{L \rightarrow \infty} R = \frac{K - m}{N - m}, \quad (2)$$

which can be adjusted by varying m , with a range of $[2R_c - 1, R_c)$, and $R_c = \frac{K}{N}$ is the code rate of component LDPC codes. It is worth noting that the staircase LDPC codes [24] are actually a special case of PS-LDPC codes when $m = \frac{N}{2}$.

B. SWD Algorithm

The overall decoding process of SWD algorithm involves two types of iterations. The inner iteration refers to decoding each component code, with a maximum of $\mathcal{I}$ iterations. The outer iteration involves message exchanges across all component codes within the window, with a maximum of $\mathcal{J}$

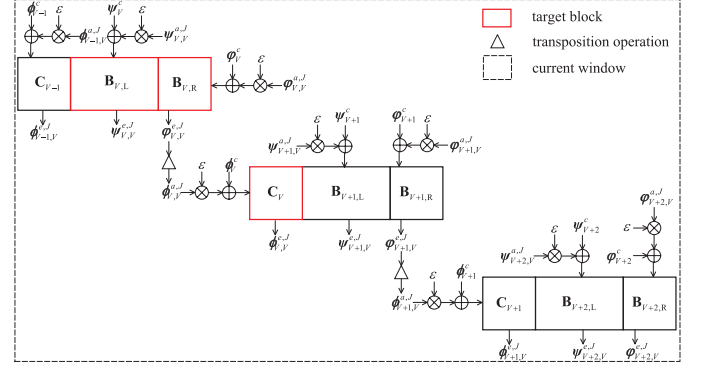


Fig. 2. The SWD process of PS-LDPC codes with target block $\mathbf{B}_V$ and $w = 4$.

iterations. In this paper, we use the BP algorithm for inner iteration, with the log-likelihood ratios (LLRs) as exchanged messages.

Let ψ_i^c , φ_i^c , and ϕ_i^c denote the channel LLRs of $\mathbf{B}_{i,L}$, $\mathbf{B}_{i,R}$, and $\mathbf{C}_i$, respectively. Similarly, let $\psi_{i,V}^{a,J}$, $\varphi_{i,V}^{a,J}$, and $\phi_{i,V}^{a,J}$ denote a priori information of $\mathbf{B}_{i,L}$, $\mathbf{B}_{i,R}$, and $\mathbf{C}_i$ in the J -th outer iteration for the target block $\mathbf{B}_V$. Correspondingly, let $\psi_{i,V}^{e,J}$, $\varphi_{i,V}^{e,J}$, and $\phi_{i,V}^{e,J}$ denote the extrinsic information of $\mathbf{B}_{i,L}$, $\mathbf{B}_{i,R}$, and $\mathbf{C}_i$ obtained after the inner iteration in the J -th outer iteration for $\mathbf{B}_V$.

The J -th outer iteration for $\mathbf{B}_V$ in SWD process of PS-LDPC codes is illustrated in Fig. 2, where w denotes the window size and ε is a scaling factor. In this process, one coupled bits matrix and $w - 1$ complete blocks are involved in the decoding of $\mathbf{B}_V$, $w \geq \Upsilon + 2$.

During the J -th outer iteration for $\mathbf{B}_V$, the input LLR matrix for the decoder of component codes is denoted as $\mathbf{L}_{i,V}^{\text{in},J}$ when decoding $\mathbf{A}_i = [\mathbf{C}_{i-1}, \mathbf{B}_{i,L}, \mathbf{B}_{i,R}]$. $\mathbf{L}_{i,V}^{\text{in},J}$ consists of channel LLRs and a priori information, $\mathbf{L}_{i,V}^{\text{in},J} = \mathbf{L}_i^c + \varepsilon \cdot \mathbf{L}_{i,V}^{a,J}$, where $\mathbf{L}_i^c = [\phi_{i-1}^c, \psi_i^c, \varphi_i^c]$, $\mathbf{L}_{i,V}^{a,J} = [\phi_{i-1,V}^{a,J}, \psi_{i,V}^{a,J}, \varphi_{i,V}^{a,J}]$, ε is used for reducing propagation of errors [24]. Based on the parallel BP decoders, $\mathbf{L}_{i,V}^{\text{in},J}$ is processed row-wise, and we can obtain the corresponding output LLR matrix $\mathbf{L}_{i,V}^{\text{out},J}$. Finally, $\mathbf{L}_{i,V}^{e,J} = [\phi_{i-1,V}^{e,J}, \psi_{i,V}^{e,J}, \varphi_{i,V}^{e,J}]$ can be obtained by $\mathbf{L}_{i,V}^{e,J} = \mathbf{L}_{i,V}^{\text{out},J} - \mathbf{L}_{i,V}^{\text{in},J}$.

Furthermore, there are three message exchanges concerning a priori information and extrinsic information during the overall decoding process as follows.

- 1) The first message exchange occurs during the initialization process after the position of current window is determined, when $V = 1$,

$$\begin{aligned} \phi_{i,V}^{a,1} &= \mathbf{0}_{m \times m}, & i &= 0, \\ \psi_{i,V}^{a,1} &= \mathbf{0}_{m \times (N-2m)}, & i &\in [1, w-1], \\ \varphi_{i,V}^{a,1} &= \mathbf{0}_{m \times m}, & i &\in [1, w-1], \end{aligned} \quad (3)$$

and when $V > 1$,

$$\begin{aligned} \phi_{i,V}^{a,1} &= \phi_{i,V-1}^{e,J*}, & i &= V-1, \\ \psi_{i,V}^{a,1} &= \begin{cases} \psi_{i,V-1}^{e,J*}, & i \in [V, V+w-3], \\ \mathbf{0}_{m \times (N-2m)}, & i = V+w-2, \end{cases} \end{aligned}$$

$$\varphi_{i,V}^{a,1} = \begin{cases} \phi_{i,V-1}^{e,J*T}, & i \in [V, V+w-3], \\ \mathbf{0}_{m \times m}, & i = V+w-2, \end{cases} \quad (4)$$

where $\mathbf{0}_{x \times y}$ is an all-zero matrix of size $x \times y$, and J^* denotes the number of outer iterations performed for decoding the previous staircase block.

- 2) The second message exchange occurs between two adjacent blocks during one outer iteration process,

$$\phi_{i,V}^{a,J} = \varphi_{i,V}^{e,JT}, i \in [V, V+w-3]. \quad (5)$$

- 3) The third message exchange occurs between two consecutive outer iteration processes, specifically at the beginning of the J -th outer iteration for $1 < J \leq \mathcal{J}$,

$$\begin{aligned} \phi_{i,V}^{a,J} &= \phi_{i,V}^{e,J-1}, & i = V-1, \\ \psi_{i,V}^{a,J} &= \psi_{i,V}^{e,J-1}, & i \in [V, V+w-2], \\ \varphi_{i,V}^{a,J} &= \begin{cases} \phi_{i,V}^{e,J-1T}, & i \in [V, V+w-3], \\ \varphi_{i,V}^{e,J-1}, & i = V+w-2. \end{cases} \end{aligned} \quad (6)$$

The target block $\mathbf{B}_V$ is involved in the decoding of both $\mathbf{A}_V$ and $\mathbf{A}_{V+1}$. To reduce error propagation, the estimated value $\hat{\mathbf{B}}_V$ of $\mathbf{B}_V$ is obtained by jointly the decoding outputs of both $\mathbf{A}_V$ and $\mathbf{A}_{V+1}$. Specifically, we first calculate $\mathbf{L}_1 = \mathbf{L}_V^c + [\phi_{V-1,V}^{e,J}, \psi_{V,V}^{e,J}, \phi_{V,V}^{e,JT}]$. Then, $\mathbf{X}_1$ and $\mathbf{X}_2$ can be obtained by hard-decision on $\mathbf{L}_1$ and $\mathbf{L}_{V+1,V}^{\text{out},J}$ at the end of each outer iteration, respectively. Next, the syndrome matrices $\mathbf{S}_1$ and $\mathbf{S}_2$ can be computed by $\mathbf{S}_1 = \mathbf{H} \cdot \mathbf{X}_1^T$ and $\mathbf{S}_2 = \mathbf{H} \cdot \mathbf{X}_2^T$, where $\mathbf{H}$ is the parity check matrix of component code. We assume that all the parity-check matrices are full-rank in this paper.

Let $\mathbf{L}(a:b)$ denote the submatrix of $\mathbf{L}$ consisting columns a to b , and $\mathbf{L}(a)$ denote the a -th column of $\mathbf{L}$. The outer iteration process is repeated until $J = \mathcal{J}$ or $\mathbf{S}_1 = \mathbf{S}_2 = \mathbf{0}_{M \times m}$, and the SWD decoder outputs $\hat{\mathbf{B}}_V = \mathbf{X}_1(m+1:N)$.

In subsequent decoding of the next target block $\mathbf{B}_{V+1}$, the decoding window slides forward by one block to incorporate a new block $\mathbf{B}_{V+w-1}$. The updated window then includes $\mathbf{C}_V, \mathbf{B}_{V+1}, \dots, \mathbf{B}_{V+w-1}$. This SWD process is repeated iteratively until $\mathbf{B}_{L-w+\Upsilon+2}$ becomes the target block. At this point, the decoding window consists of $\mathbf{C}_{L-w+\Upsilon+1}, \mathbf{B}_{L-w+\Upsilon+2}, \mathbf{B}_{L-w+\Upsilon+3}, \dots, \mathbf{B}_L, \mathbf{B}_{11}, \dots, \mathbf{B}_{1\Upsilon}$.

For target blocks indexed by $L-w+\Upsilon+2 \leq V \leq L$, the window no longer slides, and its size is dynamically reduced to $w = L + \Upsilon + 2 - V$.

C. Performance

Fig. 3 compares the block error rate (BLER) performance of PS-LDPC codes with different component codeword lengths N over the additive white Gaussian noise (AWGN) channel and binary phase shift keying (BPSK) modulation. The curve labeled ‘LDPC-Perf.’ represents the performance of corresponding component LDPC code under 1000 iterations of BP algorithm, assuming perfect a priori information is available for all coupled bits (i.e., the first m and last m bits of the component code).

First, PS-LDPC codes exhibit error floor at high E_b/N_0 , particularly for shorter N and m . Second, ‘LDPC-Perf.’ provides as an approximate lower bound on the BLER of PS-LDPC codes. Moreover, the BLER performance of PS-LDPC codes approaches that of ‘LDPC-Perf.’ for shorter

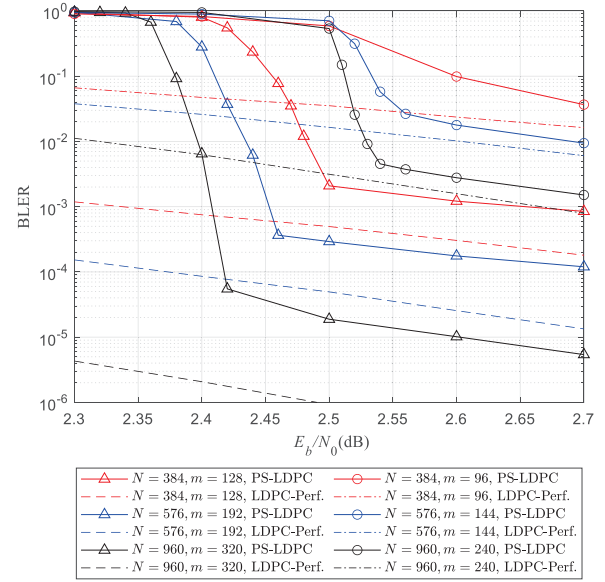


Fig. 3. The performance of SWD algorithm for PS-LDPC codes, where $R_c = 5/6$, $L = 80$, $\Upsilon = 1$, $w = 7$, $\mathcal{I} = 10$, $\mathcal{J} = 30$, $\varepsilon = 0.1$.

N or m . These observations reasonably suggest that the error floor is primarily attributed to the lack of sufficiently reliable a priori information for uncoupled bits, due to the absence of information exchange with adjacent staircase blocks. Furthermore, at a target BLER of 10^{-3} , the PS-LDPC code with $N = 576, m = 192$ performs within 0.05 dB of $N = 960, m = 320$, yet the size of block is only 36% of the latter, which can significantly reduce decoding latency. This insight motivates the decoding algorithm optimization presented in Section IV to lower the error floor.

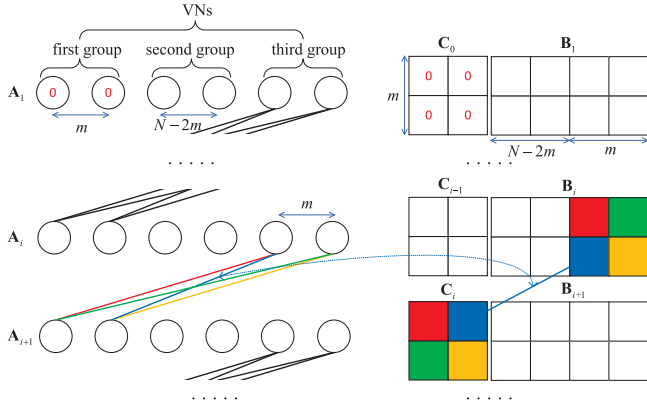
III. DENSITY EVOLUTION

In this section, we first define the node-perspective multinomials of PS-LDPC code ensemble based on the MET-LDPC framework. Then, we track the mean and variance of a priori information to derive the Gaussian approximation of DE analysis for SWD algorithm. The DE analysis adopts a block-local model with a stationarity assumption in an infinite chain. Finally, based on the above analysis, we compare the theoretical decoding thresholds with the actual BER performance of PS-LDPC codes.

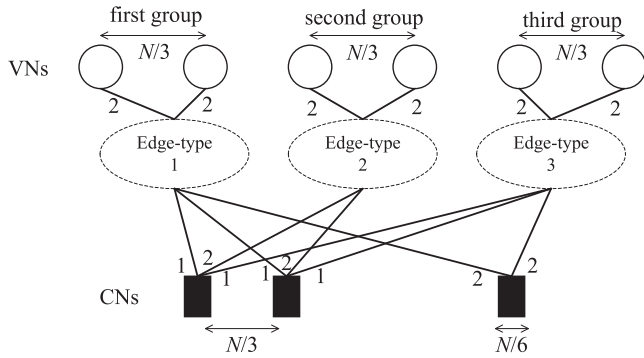
A. PS-LDPC Code Ensemble

The Tanner graph of PS-LDPC code ensemble is shown in Fig. 4, where marking edges between VNs and the corresponding bits in blocks with the same color for $\mathbf{A}_i$ and $\mathbf{A}_{i+1}$ in Fig. 4(a). From the perspective of a priori information, VNs can be divided into three different groups. The first group corresponds to $\mathbf{C}_{i-1}$, which represents the portion coupled with the previous staircase block. The second group corresponds to $\mathbf{B}_{i,L}$, which represents the uncoupled part of current block. The third group corresponds to $\mathbf{B}_{i,R}$, which represents the portion coupled with the next staircase block. Obviously, it is difficult to accurately describe the PS-LDPC code ensemble set with traditional irregular LDPC framework.

Therefore, based on the MET-LDPC framework [37], we use three edge-types to describe the PS-LDPC code ensemble



(a) Illustrations of the connectivity of VNs and the corresponding blocks, where each VN represents m bits.



(b) $\nu(x_1, x_2, x_3) = \frac{1}{3}x_1^2 + \frac{1}{3}x_2^2 + \frac{1}{3}x_3^2$ and $\tau(x_1, x_2, x_3) = \frac{1}{3}x_1x_2^2x_3 + \frac{1}{6}x_1^2x_3^2$.

Fig. 4. Tanner graph representation of a PS-LDPC code based on three edge-types with $N = 6$, $K = 3$ and $m = 2$.

where the first, second and third edge-types represent the edges connected to the first, second and third groups of VNs, respectively. We denote VNs of type- (k, d_k) and CNs of type- $\mathbf{d}$ as the VNs which connected to d_k -th edge-type, the CNs which connected to d_1 first edge-type, d_2 second edge-type and d_3 third edge-type, where $\mathbf{d} = (d_1, d_2, d_3)$, d_k is the number of edges of the k -th edge-type connected to the CNs and VNs, $k \in \{1, 2, 3\}$.

Define the node-perspective multinomials of VNs as,

$$\nu(x_1, x_2, x_3) = \sum_{d_1} \nu_{d_1} x_1^{d_1} + \sum_{d_2} \nu_{d_2} x_2^{d_2} + \sum_{d_3} \nu_{d_3} x_3^{d_3}, \quad (7)$$

and CNs as,

$$\tau(x_1, x_2, x_3) = \sum_{\mathbf{d}} \tau_{\mathbf{d}} x_1^{d_1} x_2^{d_2} x_3^{d_3}, \quad (8)$$

where ν_{d_k} is the fraction of VNs of type- (k, d_k) , and $\tau_{\mathbf{d}}$ is the fraction of CNs of type- $\mathbf{d}$ in the ensemble. In Fig. 4(b), we give an example of the Tanner graph of a PS-LDPC code, whose component code is a regular LDPC code with degree-2 VNs and degree-4 CNs.

The expressions for R_d and R_c can be derived from the node-perspective multinomials as follows,

$$R_d = 1 - \frac{\tau(1, 1, 1)}{\nu(0, 1, 1)}, \quad (9)$$

and

$$R_c = 1 - \frac{\tau(1, 1, 1)}{\nu(1, 1, 1)}. \quad (10)$$

Based on (7) and (8), the edge-perspective multinomials of VNs and CNs can be expressed as,

$$\begin{aligned} \lambda_k(x_k) &= \frac{\partial \nu(x_1, x_2, x_3) / \partial x_k}{(\partial \nu(x_1, x_2, x_3) / \partial x_k)|_{x_1=x_2=x_3=1}}, \\ &= \sum_{d_k} \lambda_k^{d_k} x_k^{d_k-1} \end{aligned} \quad (11)$$

and

$$\begin{aligned} \rho_k(x_1, x_2, x_3) &= \frac{\partial \tau(x_1, x_2, x_3) / \partial x_k}{(\partial \tau(x_1, x_2, x_3) / \partial x_k)|_{x_1=x_2=x_3=1}}, \\ &= \sum_{\mathbf{d}_k} \rho_k^{\mathbf{d}_k} x_k^{d_k-1} x_j^{d_j} x_l^{d_l} \end{aligned} \quad (12)$$

respectively, where $j \in \{1, 2, 3\} \setminus \{k\}$, $l \in \{1, 2, 3\} \setminus \{k, j\}$, $\lambda_k^{d_k} = \frac{d_k \nu_{d_k}}{\sum_{d_k} d_k \nu_{d_k}}$ denote the fraction of k -th edge-type connected to VNs of type- (k, d_k) , and $\rho_k^{\mathbf{d}_k} = \frac{d_k \tau_{\mathbf{d}_k}}{\sum_{\mathbf{d}_k} d_k \tau_{\mathbf{d}_k}}$ denote the fraction of k -th edge-type connected to CNs of type- $\mathbf{d}$. For the PS-LDPC code shown in Fig. 4(b), the edge-perspective multinomials are given by $\lambda_1(x_1) = x_1$, $\lambda_2(x_2) = x_2$, $\lambda_3(x_3) = x_3$, $\rho_1(x_1, x_2, x_3) = \frac{1}{2}x_2^2x_3 + \frac{1}{2}x_1x_3^2$, $\rho_2(x_1, x_2, x_3) = x_1x_2x_3$, $\rho_3(x_1, x_2, x_3) = \frac{1}{2}x_1x_2^2 + \frac{1}{2}x_1^2x_3$.

B. DE Approximation Analysis

Due to the high complexity of exact DE, Gaussian approximation [43] is generally adopted in practical analysis. Generally, the iterative messages of BP decoding satisfy the symmetry condition [43]: $h(x) = e^x h(-x)$, where $h(x)$ is the PDF of variable x . This condition implies that the variance of the iterative messages is as twice of its mean. Therefore, the evolution of PDF of iterative messages can be described by tracking the changes of the mean, which is the so-called single-parameter Gaussian approximation. However, in SWD algorithm, ε leads to the symmetry condition no longer being satisfied. Thus, the single-parameter Gaussian approximation is not applicable to SWD algorithm of PS-LDPC codes.

Without loss of generality, we consider the transmitter to send an all-zero signal over the AWGN channel with BPSK modulation, and define σ_0^2 as noise variance. Therefore, ψ_i^c , φ_i^c and ϕ_i^c follow normal distribution $\mathcal{N}(\frac{2}{\sigma_0^2}, \frac{4}{\sigma_0^2})$. Denote $f_{\mathcal{N}}(x; \mu, \sigma^2)$ as probability density function (PDF) of normal distribution $\mathcal{N}(\mu, \sigma^2)$, then the PDF of $\mathbf{L}_i^c$ is

$$f_c(x) = f_{\mathcal{N}}(x; \mu_c, \sigma_c^2), \quad (13)$$

where $\mu_c = \frac{2}{\sigma_0^2}$ and $\sigma_c^2 = \frac{4}{\sigma_0^2}$.

Furthermore, denote the PDF of $\psi_{i,V}^{a,J}$, $\varphi_{i,V}^{a,J}$, $\phi_{i,V}^{a,J}$, $\psi_{i,V}^{e,J}$, $\varphi_{i,V}^{e,J}$ and $\phi_{i,V}^{e,J}$ as $f_{\psi,i,V,J}(x)$, $f_{\varphi,i,V,J}(x)$, $f_{\phi,i,V,J}(x)$, $g_{\psi,i,V,J}(x)$, $g_{\varphi,i,V,J}(x)$ and $g_{\phi,i,V,J}(x)$, respectively. To simplify, we omit the subscript V and J in these PDFs. Assuming that the VNs follow Gaussian distribution, and after the inner iteration, the distribution of extrinsic information of VNs with the same degree in the block is identical, then a priori

TABLE I
THE VALUES OF a , b , AND c IN DIFFERENT CASES FOR $w > 3$

Case		a	b	c
$V = 1$	$J = 1$	$i = V$	-	-
		$V < i \leq V + w - 2$	3	-
	$J > 1$	$i = V$	1	2
		$V < i < V + w - 2$	3	2
		$i = V + w - 2$	3	2
$V > 1$	$J = 1$	$i = V$	1	2
		$V < i < V + w - 3$	3	2
		$i = V + w - 3$	3	2
		$i = V + w - 2$	3	-
	$J > 1$	$i = V$	1	2
		$V < i < V + w - 2$	3	2
		$i = V + w - 2$	3	2

information follows mixture Gaussian distribution. Thus, these PDFs of a priori information can be expressed as,

$$f_{\psi,i}(x) = \sum_{d_b} \frac{\nu_{d_b} f_{\mathcal{N}}(x; \mu_{\psi,d_b,i}, \sigma_{\psi,d_b,i}^2)}{\sum_{d_b} \nu_{d_b}}, \quad (14)$$

$$f_{\varphi,i}(x) = \sum_{d_c} \frac{\nu_{d_c} f_{\mathcal{N}}(x; \mu_{\varphi,d_c,i}, \sigma_{\varphi,d_c,i}^2)}{\sum_{d_c} \nu_{d_c}}, \quad (15)$$

and

$$f_{\phi,i-1}(x) = \sum_{d_a} \frac{\nu_{d_a} f_{\mathcal{N}}(x; \mu_{\phi,d_a,i-1}, \sigma_{\phi,d_a,i-1}^2)}{\sum_{d_a} \nu_{d_a}}, \quad (16)$$

respectively, where the values of a , b , and c in different cases are shown in Table I, $\mu_{\psi,d_k,i}$, $\mu_{\varphi,d_k,i}$ and $\mu_{\phi,d_k,i-1}$ denote the mean of VNs of type- (k, d_k) for $\psi_{i,V}^{a,J}$, $\varphi_{i,V}^{a,J}$ and $\phi_{i-1,V}^{a,J}$, respectively, $\sigma_{\psi,d_k,i}^2$, $\sigma_{\varphi,d_k,i}^2$ and $\sigma_{\phi,d_k,i-1}^2$ denote the variance of VNs of type- (k, d_k) for $\psi_{i,V}^{a,J}$, $\varphi_{i,V}^{a,J}$ and $\phi_{i-1,V}^{a,J}$, respectively.

Based on Eqs. (13)-(16), the PDF of $\mathbf{L}_{i,V}^{\text{in},J}$ can be expressed as,

$$f_i^0(x) = \sum \nu_{d_b} f_{\mathcal{N}}(x; \mu_c + \varepsilon \mu_{\psi,d_b,i}, \sigma_c^2 + \varepsilon^2 \sigma_{\psi,d_b,i}^2) + \nu_{d_c} f_{\mathcal{N}}(x; \mu_c + \varepsilon \mu_{\varphi,d_c,i}, \sigma_c^2 + \varepsilon^2 \sigma_{\varphi,d_c,i}^2) + \nu_{d_a} f_{\mathcal{N}}(x; \mu_c + \varepsilon \mu_{\phi,d_a,i-1}, \sigma_c^2 + \varepsilon^2 \sigma_{\phi,d_a,i-1}^2). \quad (17)$$

Then, we define the PDF of variable-to-check messages from VNs along the k -th edge-type at the I -th inner iteration as $f_{i,k}^I(x)$ for decoding $\mathbf{A}_i$ in the J -th outer iteration when the target block is $\mathbf{B}_V$. The subscript i is omitted for simplify, and we have

$$f_k^I(x) = \sum_{d_k} \frac{\lambda_k^{\nu_{d_k}} \nu_{d_k}}{\sum_{d_k} \nu_{d_k}} f_{\mathcal{N}}(x; \check{\mu}_{d_k,I}, \check{\sigma}_{d_k,I}^2), \quad (18)$$

where $z = a$ for $k = 1$, $z = b$ for $k = 2$, $z = c$ for $k = 3$, $\check{\mu}_{d_k,I}$ and $\check{\sigma}_{d_k,I}^2$ are the mean and variance of variable-to-check

messages from VNs of type- (k, d_k) , respectively, and can be given by,

$$\check{\mu}_{d_k,I} = \begin{cases} \mu_c + \varepsilon \mu_{\phi,d_a,i-1} + (d_1 - 1) \hat{\mu}_{1,I-1}, & k = 1, \\ \mu_c + \varepsilon \mu_{\psi,d_b,i} + (d_2 - 1) \hat{\mu}_{2,I-1}, & k = 2, \\ \mu_c + \varepsilon \mu_{\varphi,d_c,i} + (d_3 - 1) \hat{\mu}_{3,I-1}, & k = 3, \end{cases} \quad (19)$$

and

$$\check{\sigma}_{d_k,I}^2 = \begin{cases} \sigma_c^2 + \varepsilon^2 \sigma_{\phi,d_a,i-1}^2 + (d_1 - 1) \hat{\sigma}_{1,I-1}^2, & k = 1, \\ \sigma_c^2 + \varepsilon^2 \sigma_{\psi,d_b,i}^2 + (d_2 - 1) \hat{\sigma}_{2,I-1}^2, & k = 2, \\ \sigma_c^2 + \varepsilon^2 \sigma_{\varphi,d_c,i}^2 + (d_3 - 1) \hat{\sigma}_{3,I-1}^2, & k = 3, \end{cases} \quad (20)$$

where $\hat{\mu}_{k,I-1}$ and $\hat{\sigma}_{k,I-1}^2$ denote the mean and variance of the check-to-variable messages from CNs along the k -th edge-type at the $(I - 1)$ -th inner iteration, respectively, and we have

$$\hat{\mu}_{k,I} = \sum_d \rho_k^d \hat{\mu}_{d,k,I}, \quad (21)$$

and

$$\hat{\sigma}_{k,I}^2 = \sum_d \rho_k^d \hat{\sigma}_{d,k,I}^2, \quad (22)$$

where $\hat{\mu}_{d,k,I}$ and $\hat{\sigma}_{d,k,I}^2$ denote the mean and variance of the check-to-variable messages from CNs of type- d along the k -th edge-type at the I -th inner iteration, respectively, and can be given by,

$$\hat{\mu}_{d,k,I} = \int \dots \int 2 \tanh^{-1} \left(\prod_{i=1}^{d'-1} \tanh \left(\frac{x_i}{2} \right) \right) \prod_{i=1}^{d_k-1} f_k^I(x_i) \prod_{i=d_k+d_j-1}^{d_k+d_j-1} f_j^I(x_i) \prod_{i=d_k+d_j}^{d'-1} f_l^I(x_i) dx_1 \dots dx_{d'-1}, \quad (23)$$

and

$$\hat{\sigma}_{d,k,I}^2 = \int \dots \int \left(2 \tanh^{-1} \left(\prod_{i=1}^{d'-1} \tanh \left(\frac{x_i}{2} \right) \right) \right)^2 \prod_{i=1}^{d_k-1} f_k^I(x_i) \prod_{i=d_k+d_j-1}^{d_k+d_j-1} f_j^I(x_i) \prod_{i=d_k+d_j}^{d'-1} f_l^I(x_i) dx_1 \dots dx_{d'-1} - \hat{\mu}_{d,k,I}^2, \quad (24)$$

where $j \in \{1, 2, 3\} \setminus \{k\}$, $l \in \{1, 2, 3\} \setminus \{k, j\}$, $d' = d_1 + d_2 + d_3$. The multiple definite integrals above can be calculated by Monte Carlo method.

When $I = \mathcal{I}$, the inner iteration ends and $g_{\psi,i}(x)$, $g_{\varphi,i}(x)$ and $g_{\phi,i-1}(x)$ can be further expressed as,

$$g_{\psi,i}(x) = \sum_{d_2} \frac{\nu_{d_2} f_{\mathcal{N}}(x; d_2 \hat{\mu}_{2,\mathcal{I}}, d_2 \hat{\sigma}_{2,\mathcal{I}}^2)}{\sum_{d_2} \nu_{d_2}}, \quad (25)$$

$$g_{\varphi,i}(x) = \sum_{d_3} \frac{\nu_{d_3} f_{\mathcal{N}}(x; d_3 \hat{\mu}_{3,\mathcal{I}}, d_3 \hat{\sigma}_{3,\mathcal{I}}^2)}{\sum_{d_3} \nu_{d_3}}, \quad (26)$$

$$g(x) = \sum \frac{\nu_{d_2}}{\sum \nu_{d_2} + \nu_{d_1}} f_{\mathcal{N}}(x; \mu_c + \varepsilon \mu_{\phi,d_2,V} + d_2 \hat{\mu}_{2,\mathcal{I}}, \sigma_c^2 + \varepsilon^2 \sigma_{\phi,d_2,V}^2 + d_2 \hat{\sigma}_{2,\mathcal{I}}^2) + \frac{\nu_{d_3} \nu_{d_1}}{(\sum \nu_{d_2} + \nu_{d_1}) \sum \nu_{d_3}} f_{\mathcal{N}}(x; \mu_c + \varepsilon \mu_{\psi,d_3,V+1} + d_1 \hat{\mu}_{1,\mathcal{I}}, \sigma_c^2 + \varepsilon^2 \sigma_{\psi,d_3,V+1}^2 + d_1 \hat{\sigma}_{1,\mathcal{I}}^2). \quad (28)$$

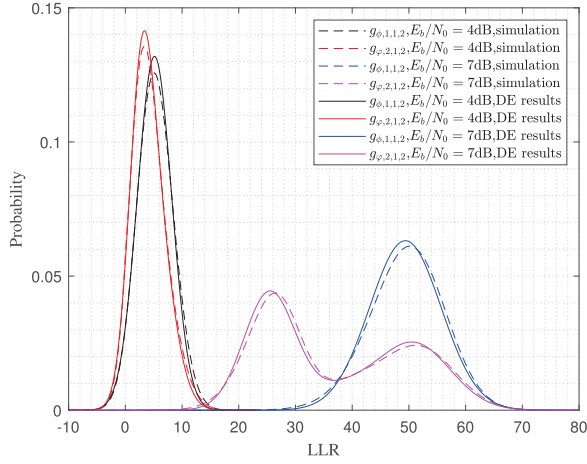
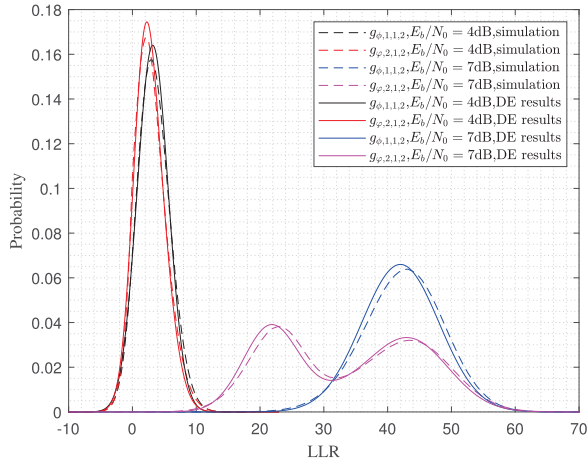
(a) $m = 768$.(b) $m = 1152$.

Fig. 5. Simulation results of the LLR function with SWD algorithm and DE analysis, where $N = 2304$, $K = 1728$, $\mathcal{I} = 1$, $\mathcal{J} = 2$, $w = 4$, $\varepsilon = 0.5$. (a) $\nu(x_1, x_2, x_3) = \frac{1}{3}x_1^4 + \frac{1}{3}x_2^4 + \frac{5}{24}x_3^2 + \frac{1}{24}x_3^3 + \frac{1}{12}x_3^4$, $\tau(x_1, x_2, x_3) = \frac{1}{24}x_1^4x_2^6x_3^4 + \frac{1}{24}x_1^4x_2^8x_3^2 + \frac{1}{24}x_1^2x_2^6x_3^3 + \frac{1}{24}x_1^6x_2^4x_3^3 + \frac{1}{24}x_1^6x_2^2x_3^5$, and (b) $\nu(x_1, x_2, x_3) = \frac{1}{5}x_1^4 + \frac{5}{24}x_2^2 + \frac{1}{24}x_3^3 + \frac{1}{4}x_3^4$, $\tau(x_1, x_2, x_3) = \frac{1}{12}x_1^7x_3^7 + \frac{3}{24}x_1^8x_3^6 + \frac{1}{24}x_1^{10}x_3^5$.

and

$$g_{\phi,i-1}(x) = \sum_{d_1} \frac{\nu_{d_1} f_N(x; d_1 \hat{\mu}_{1,\mathcal{I}}, d_1 \hat{\sigma}_{1,\mathcal{I}}^2)}{\sum_{d_1} \nu_{d_1}}, \quad (27)$$

respectively. Fig. 5 compares the LLR function with SWD algorithm and DE analysis for different m , we can observe that the DE analysis results agree well with the decoding results, which validate the accuracy of our Gaussian approximation analysis.

C. Threshold

Based on the above Gaussian approximation analysis, the PDF of $\mathbf{L}_1(m+1 : N)$ are given by Eq. (28), shown at the bottom of the previous page. For a target error probability δ , the decoding threshold is defined as $\sigma_{\text{th}} = \sup \left\{ \sigma_0 > 0 \mid \lim_{J \rightarrow \infty} \int_{-\infty}^0 g(x) dx \leq \delta, \exists I \in \mathbb{Z}^+, \exists \varepsilon > 0, \forall V \in [1, \dots, L] \right\}$. Fig. 6 compares the thresholds and BER

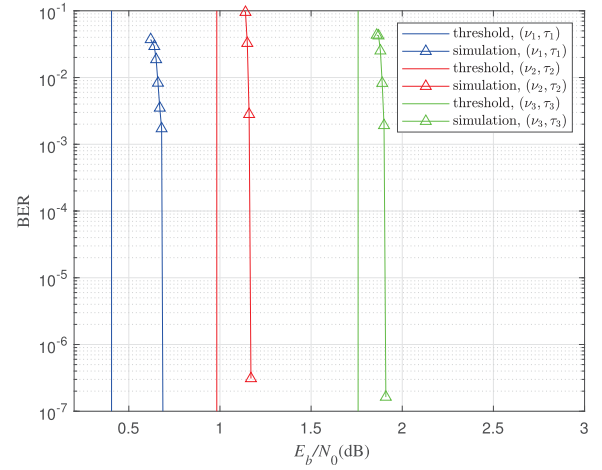


Fig. 6. BER vs E_b/N_0 thresholds of PS-LDPC codes with $N = 2304$, $L = 40$, $\Upsilon = 1$, $w = 4$, $\mathcal{I} = 20$, $\mathcal{J} = 20$, $\varepsilon = 0.5$, where $\nu_1(x_1, x_2, x_3) = \frac{1}{2}x_1^4 + \frac{7}{24}x_3^2 + \frac{1}{24}x_3^3 + \frac{1}{6}x_3^4$, $\tau_1(x_1, x_2, x_3) = \frac{7}{24}x_1^4x_3^4 + \frac{1}{24}x_1^6x_3^5$, $\nu_2(x_1, x_2, x_3) = \frac{1}{2}x_1^4 + \frac{5}{24}x_2^2 + \frac{1}{24}x_3^3 + \frac{1}{4}x_3^4$, $\tau_2(x_1, x_2, x_3) = \frac{1}{12}x_1^7x_3^7 + \frac{1}{8}x_1^8x_3^6 + \frac{1}{24}x_1^{10}x_3^5$, $\nu_3(x_1, x_2, x_3) = \frac{3}{8}x_1^3 + \frac{1}{8}x_1^4 + \frac{1}{8}x_2^2 + \frac{1}{24}x_3^3 + \frac{1}{3}x_3^4$, $\tau_3(x_1, x_2, x_3) = \frac{3}{24}x_1^{10}x_3^{10} + \frac{1}{24}x_1^9x_3^{11}$.

performance of three PS-LDPC codes, we can observe that the BER curves are in accordance with their corresponding thresholds.

IV. ALGORITHM OPTIMIZATION

In this section, we first optimize the coupling patterns of PS-LDPC codes by designing the OCP encoding algorithm. Then, we propose the C-SWD algorithm to improve decoding performance and lower the error floor. Finally, a comprehensive computational complexity analysis is provided for our PS-LDPC coding scheme and compare to the related coding schemes [30], [31], [32].

A. Coupling Patterns Optimization

Our analysis shows that even with identical component LDPC codes, different coupled bits positions (i.e., coupling patterns) results in distinct PS-LDPC code ensembles and affect performance. To optimize the coupling patterns, we employ the DE approximation framework presented in Section III. This allows us to evaluate the decoding thresholds for various coupling patterns, and identify the optimal configuration that minimizes the threshold.

The proposed encoding algorithm is shown in Algorithm 1, where the column permutations Π_1 and Π_2 are determined based on DE approximation analysis to ensure the optimal coupling patterns. The first column permutation Π_1 is used to select m VNs from the first K VNs of component code as the first group of VNs in PS-LDPC code. The second column permutation Π_2 is applied after the first group of VNs is determined, and it divides the remaining $N - K$ VNs of component code into the second and third groups of VNs in PS-LDPC code. The resulting coupling does not affect the inherent properties of the component code itself. Therefore, using quasi-cyclic (QC)-LDPC codes as component codes is suitable for high-speed signal processing. Exhaustively searching for the optimal coupling pattern is computationally

Algorithm 1 OCP Encoding Algorithm**Input:** $\mathbf{B}_{1,\text{info}}, \mathbf{B}_{2,\text{info}}, \dots, \mathbf{B}_{L,\text{info}}, \mathbf{G}, m, \Pi_1, \Pi_2, \Upsilon$ **Output:** $\mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_L, \mathbf{B}_{t1}, \mathbf{B}_{t2}, \dots, \mathbf{B}_{t\Upsilon}$

- 1: Initialize $m \times m$ matrix $\mathbf{B}_0$ as all zeros
- 2: **for** $i = 1, 2, \dots, L$ **do**
- 3: Obtain $\mathbf{C}_{i-1} = \mathbf{B}_{i-1,\text{R}}^T$, where
 $\mathbf{B}_{i-1,\text{R}} = \mathbf{B}_{i-1}(N - 2m + 1 : N - m)$.
- 4: Perform the first permutation Π_1 : $m \times K$ matrix $\bar{\mathbf{B}}_i = \Pi_1([\mathbf{C}_{i-1}, \mathbf{B}_{i,\text{info}}])$
- 5: Encoding row by row, $\mathbf{B}_{i,\text{parity}} = \bar{\mathbf{B}}_i \cdot \mathbf{G}$
- 6: Obtain $\mathbf{B}_i = [\mathbf{B}_{i,\text{info}}, \mathbf{B}_{i,\text{parity}}]$
- 7: Perform the second permutation Π_2 : $\mathbf{B}_i = \Pi_2(\mathbf{B}_i)$
- 8: **end for**
- 9: Zero-padding and repeat 3-7 to obtain termination blocks,
 $\mathbf{B}_{t1}, \mathbf{B}_{t2}, \dots, \mathbf{B}_{t\Upsilon}$
- 10: **return** $\mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_L, \mathbf{B}_{t1}, \mathbf{B}_{t2}, \dots, \mathbf{B}_{t\Upsilon}$

prohibitive. For the protograph LDPC codes, instead of performing permutations at bit level, we search over permutations at the level of lifting factor Z to obtain a suboptimal coupling pattern in this paper.

Example 1: As an illustration, we select $\mathcal{C}(960, 800)$ specified in IEEE 802.16e standard as the component LDPC code, where $Z = 40$. For $m = 320$, the conventional encoding process, where the first group of VNs consists of the first 320 bits, the second group consists of bits 321 to 640, and the third group consists of the last 320 bits, corresponds to the PS-LDPC code ensemble with $\nu(x_1, x_2, x_3) = \frac{1}{3}x_1^3 + \frac{1}{24}x_2^3 + \frac{7}{24}x_2^4 + \frac{1}{8}x_3^2 + \frac{1}{24}x_3^3 + \frac{1}{6}x_3^4$, $\tau(x_1, x_2, x_3) = \frac{1}{8}x_1^6x_2^8x_3^6 + \frac{1}{24}x_1^6x_2^7x_3^7$ and $\sigma_{\text{th}} = 0.6521$. After Π_1 and Π_2 , the first group of VNs consists of bits 401 to 720, the second group consists of bits 1 to 200, 241 to 280, and 841 to 920, and the third group consists the remaining bits. In this case, $\nu(x_1, x_2, x_3) = \frac{1}{3}x_1^4 + \frac{1}{12}x_2^2 + \frac{1}{4}x_2^3 + \frac{1}{24}x_3^2 + \frac{1}{6}x_3^3 + \frac{1}{8}x_3^4$, $\tau(x_1, x_2, x_3) = \frac{1}{24}x_1^8x_2^5x_3^3 + \frac{1}{24}x_1^8x_2^6x_3^3 + \frac{1}{24}x_1^8x_2^5x_3^5 + \frac{1}{24}x_1^8x_2^4x_3^8$, and $\sigma_{\text{th}} = 0.6615$.

Based on the proposed encoding algorithm, the message exchange process and the inner iteration in SWD algorithm also requires corresponding modifications. The input to the component decoder is modified as,

$$\bar{\mathbf{L}}_{i,V}^{\text{in},J} = [\Pi_1[\mathbf{L}_{i,V}^{\text{in},J}(1:K)], \mathbf{L}_{i,V}^{\text{in},J}(K+1:N)], \quad (29)$$

and $\mathbf{L}_{i,V}^{\text{out},J}$ is obtained by

$$\mathbf{L}_{i,V}^{\text{out},J} = [\Pi_1^{-1}[\bar{\mathbf{L}}_{i,V}^{\text{out},J}(1:K)], \bar{\mathbf{L}}_{i,V}^{\text{out},J}(K+1:N)], \quad (30)$$

where $\bar{\mathbf{L}}_{i,V}^{\text{out},J}$ denotes the output LLR matrix of decoder.

Moreover, Eqs. (4)-(6) are modified as Eqs. (31)-(33) as follows

$$\begin{aligned}
 \phi_{i,V}^{a,1} &= \phi_{i,V-1}^{e,J*}, \quad i = V-1, \\
 \Phi_{i,V}^{a,1}(1:N-2m) &= \begin{cases} \Phi_{i,V-1}^{e,J*}(1:N-2m), & i \in [V, V+w-3], \\ \mathbf{0}_{m \times (N-2m)}, & i = V+w-2, \end{cases} \\
 \Phi_{i,V}^{a,1}(N-2m+1:N-m) &= \begin{cases} \phi_{i,V-1}^{e,J*T}, & i \in [V, V+w-3], \\ \mathbf{0}_{m \times m}, & i = V+w-2. \end{cases} \quad (31)
 \end{aligned}$$

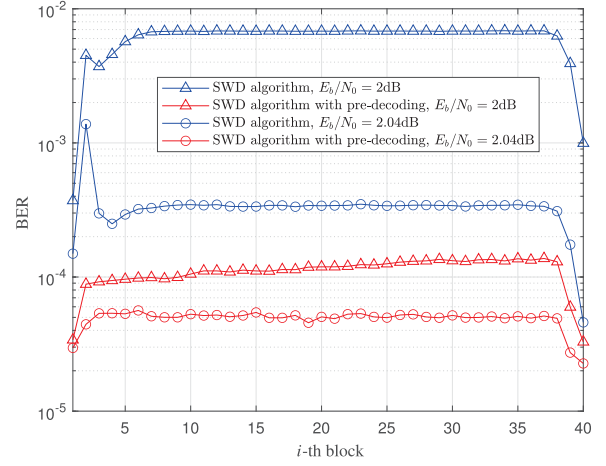


Fig. 7. BER of the i -th block in PS-LDPC codes, where $N = 384$, $K = 288$, $m = 128$, $L = 40$, $\Upsilon = 1$, $w = 5$, $\mathcal{I} = 10$, $\mathcal{J} = 10$, $\varepsilon = 0.1$.

where $\Phi_{i,V}^{a,J} = \Pi_2[\psi_{i,V}^{a,J}, \varphi_{i,V}^{a,J}]$, $\Phi_{i,V}^{e,J} = \Pi_2[\psi_{i,V}^{e,J}, \varphi_{i,V}^{e,J}]$, and

$$\Phi_{i,V}^{a,JT} = \Phi_{i,V}^{e,J}(N-2m+1:N-m), \quad i \in [V, V+w-3], \quad (32)$$

and

$$\begin{aligned}
 \phi_{i,V}^{a,J} &= \phi_{i,V}^{e,J-1}, \quad i = V-1, \\
 \Phi_{i,V}^{a,J}(1:N-2m) &= \Phi_{i,V}^{e,J-1}(1:N-2m), \quad i \in [V, V+w-3], \\
 \Phi_{i,V}^{a,J} &= \Phi_{i,V}^{e,J-1}, \quad i = V+w-2. \quad (33)
 \end{aligned}$$

B. C-SWD Algorithm

In this subsection, we propose the C-SWD algorithm by modifying the SWD algorithm to accommodate PS-LDPC codes. The algorithm primarily improves performance by optimizing the decoding process of the initial block, enhancing the reliability of coupled bits based on the syndrome matrices, and cascading OLD or BP decoders, primarily to correct errors in non-coupled bits.

Firstly, we introduce a pre-decoding step to enhance performance by increasing the decoding success probability of the initial blocks. The pre-decoding is an additional decoding step performed for $\mathbf{B}_1$ before the SWD process, which has the window size of $w-1$. Then, $\mathbf{B}_1$ is decoded by the SWD with the window size of w again. Specifically, it decodes the window comprising $\mathbf{C}_0, \mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_{w-2}$ until $J = \mathcal{J}$ or $\mathbf{S}_1 = \mathbf{S}_2 = \mathbf{0}_{M \times m}$. The extrinsic information of each block obtained from this pre-decoding step is used for initialization process of SWD algorithm. The performance of SWD algorithm with pre-decoding is shown in Fig. 7. It can be observed that the BER of $\mathbf{B}_2$ in SWD algorithm is significantly higher than that of the other blocks for $E_b/N_0 = 2.04$ dB. This is due to the early termination of iterations during the decoding of $\mathbf{B}_1$, which prevents sufficient iteration to update a priori information. The introduction of pre-decoding effectively alleviates this issue and improves overall decoding performance.

Secondly, after outputting $\tilde{\mathbf{B}}_V, \phi_V^c$ and $\phi_{V,V}^{e,J*}$ are modified based on $\mathbf{S}_1$ and $\mathbf{S}_2$. Let $\mathbf{L} = |\mathbf{L}| \cdot \text{sign}(\mathbf{L})$, where $|\mathbf{L}|$ and

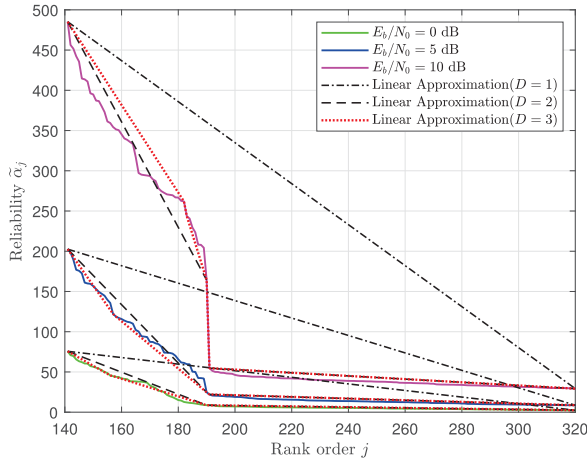


Fig. 8. Reliability curve related to the MRB for cascaded OLD, $\kappa = 10$.

$\text{sign}(\mathbf{L})$ represent the element-wise absolute value and sign matrices of $\mathbf{L}$, respectively. If $\mathbf{S}_1 = \mathbf{S}_2 = \mathbf{0}_{M \times m}$, $\mathbf{C}_V$ satisfies double parity-check constraints, which indicates that its error probability is extremely low. Thus, we have

$$\begin{aligned} |\phi_V^c| &= \infty, & \text{sign}(\phi_V^c) &= (-1)^{\mathbf{X}_2(1:m)}, \\ |\phi_{V,V}^{e,J*}| &= \infty, & \text{sign}(\phi_{V,V}^{e,J*}) &= (-1)^{\mathbf{X}_2(1:m)}. \end{aligned} \quad (34)$$

If $J = \mathcal{J}$ and $\mathbf{S}_2 \neq \mathbf{0}_{M \times m}$, some bits in $\mathbf{C}_V$ may still satisfy the single parity-check constraint. Therefore, a reliability factor $\kappa > 1$ is introduced, and for $\mathbf{S}_1(i) = \mathbf{0}_{M \times 1}$, $1 \leq i \leq m$, we have

$$\begin{aligned} |\phi_V^c(i)| &= \kappa \cdot |\phi_V^c(i)|, & \text{sign}(\phi_V^c(i)) &= (-1)^{\mathbf{X}_2(i)}, \\ |\phi_{V,V}^{e,\mathcal{J}}(i)| &= \kappa \cdot |\phi_{V,V}^{e,\mathcal{J}}(i)|, & \text{sign}(\phi_{V,V}^{e,\mathcal{J}}(i)) &= (-1)^{\mathbf{X}_2(i)}. \end{aligned} \quad (35)$$

In general, optimal κ tends to be proportional to N and inversely proportional to the minimum distance of component code. Based on our simulations, $10 \leq \kappa \leq 50$ works effectively.

Thirdly, set the constant T_{flag} . It is ensured that the cascaded decoders are not attempted in cases where the complexity of the worst case is too high. Based on our empirical observations, a suitable value of T_{flag} generally falls within the range of $0.1m$ to $0.4m$. When $J = \mathcal{J}$ and $0 < t_S \leq T_{\text{flag}}$, perform cascaded decoders on the rows of $\mathbf{A}_V$ that do not satisfy the parity-check constraint, where t_S is the number of non-zero columns in $\mathbf{S}_1$. Specifically, we can predict the error floor of PS-LDPC codes by the performance of ‘LDPC-Perf.’ in Section II-C, and the threshold in Section III-C. If the performance of ‘LDPC-Perf.’ meets the reliability requirements, the BP decoder can be employed to achieve this performance. Otherwise, the OLD is used to lower the error floor.

For $\mathbf{S}_1(i) \neq \mathbf{0}_{M \times 1}$, the input to the cascaded decoder is the i -th row of $\mathbf{L}_V^c$, $\mathbf{l} = [l_1, l_2, \dots, l_N]$, to avoid the impact of erroneous a prior information. Before decoding, if $\mathbf{S}_2(j) = \mathbf{0}_{M \times 1}$, we have

$$l_{j+N-m} = \kappa \cdot |l_{j+N-m}| \cdot (-1)^{x_j}, \quad (36)$$

where x_j is the i -th element of $\mathbf{X}_1(j+N-m)$. The decoding results that satisfy the parity-check constraints are used to update $\phi_V^c(i)$ and $\phi_{V,V}^{e,\mathcal{J}}(i)$ by Eq.(35).

Algorithm 2 C-SWD Algorithm

Input: Channel LLRs, L , Υ , $\mathbf{H}$, m , $\mathcal{I}$, $\mathcal{J}$, Π_1 , Π_2 , w , ε , κ , t_{flag}

Output: Optimal estimation $\tilde{\mathbf{B}}_1, \tilde{\mathbf{B}}_2, \dots, \tilde{\mathbf{B}}_L$

- 1: Perform pre-decoding to obtain the extrinsic information of $\mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_{w-2}$
- 2: **for** $V = 1, 2, \dots, L - w + \Upsilon + 1$ **do**
- 3: First message exchange by Eq.(31) or pre-decoding
- 4: **for** $J = 1, 2, \dots, \mathcal{J}$ **do**
- 5: Third message exchange based on Eq.(33)
- 6: **for** $t = V, V + 1, \dots, V + w - 2$ **do**
- 7: Calculate $\tilde{\mathbf{L}}_{t,V}^{\text{in},J}$ by Eq.(29)
- 8: Obtain $\tilde{\mathbf{L}}_{t,V}^{\text{out},J}$ through inner iteration
- 9: Calculate $\mathbf{L}_{t,V}^{\text{out},J}$ by Eq.(30)
- 10: Second message exchange based on Eq.(32)
- 11: **end for**
- 12: Calculate $\mathbf{S}_1$ and $\mathbf{S}_2$
- 13: **if** $\mathbf{S}_1 = \mathbf{S}_2 = \mathbf{0}_{M \times m}$ **then**
- 14: Stop outer iteration and output $\tilde{\mathbf{B}}_V$
- 15: Modify ϕ_V^c and $\phi_{V,V}^{e,J*}$ by Eq.(34)
- 16: **else if** $J = \mathcal{J}$ **then**
- 17: **else if** $0 < t_S \leq T_{\text{flag}}$ **then**
- 18: Run the cascaded decoders based on Eq.(36)
- 19: Output $\tilde{\mathbf{B}}_V$
- 20: Modify ϕ_V^c and $\phi_{V,V}^{e,\mathcal{J}}$ by Eq.(35)
- 21: **else**
- 22: Output $\tilde{\mathbf{B}}_V$
- 23: Modify ϕ_V^c and $\phi_{V,V}^{e,\mathcal{J}}$ by Eq.(35)
- 24: **end if**
- 25: **end for**
- 26: **end for**
- 27: **for** $V = L - w + \Upsilon + 2, \dots, L$ **do**
- 28: Adjust the window size and repeat 3-25
- 29: **end for**
- 30: **return** $\tilde{\mathbf{B}}_1, \tilde{\mathbf{B}}_2, \dots, \tilde{\mathbf{B}}_L$

The key of OLD lies in its TEP generation criterion [40], [44], where the reliability sequence related to MRB is approximated with multiple lines of segment number $D \geq 1$. Specifically, the reliability sequence is firstly computed as $\alpha = |\mathbf{l}|$. Next, sort α in descending order and obtain the permutation Π_3 . Apply Π_3 to $\mathbf{G}$ to get $\tilde{\mathbf{G}} = \Pi_3(\mathbf{G})$. Then, perform Gaussian elimination on $\tilde{\mathbf{G}}$ to obtain $\tilde{\mathbf{G}}$ and the permutation Π_4 . Thus, $\tilde{\alpha} = \Pi_4(\Pi_3(\alpha)) = (\tilde{\alpha}_1, \tilde{\alpha}_2, \dots, \tilde{\alpha}_N)$, and the reliability sequence related to MRB is $\tilde{\alpha}_B = (\tilde{\alpha}_1, \tilde{\alpha}_2, \dots, \tilde{\alpha}_K)$. Assuming that the reliability has been enhanced after decoding the previous block according to Eq.(34), the first m bits of MRB no longer require error correction, and the length of TEP can be reduced from K to $(K - m)$.

As shown in Fig. 8, we plot the reliability curve of $\tilde{\alpha}_B$ for the cascaded OLD in single realization of component code $\mathcal{C}(384, 320)$, where $m = 140$ bits are modified by Eq.(34), and 50 bits are modified by Eq.(36). We can observe that for the cascaded OLD, the number of segments D should be at least 2 to ensure a better match between the fitted lines and the reliability curve. Notably, the output of cascaded OLD always satisfies parity-check constraint. To avoid error propagation

and reduce complexity, an additional condition $t'_S \leq T'_{\text{flag}}$ is imposed, where t'_S is the number of non-zero columns in $\mathbf{S}_2$. The cascaded OLD is activated only when this condition holds.

The PS-SWD algorithm is summarized in Algorithm 2.

C. Computational Complexity

In this subsection, we analyze the decoding complexity of our PS-LDPC coding scheme and other related coding schemes by comparing the average number of operations, which is defined as the total number of real additions, binary additions, multiplications, comparisons and table look-ups.

We first derive the computational complexity of a single BP decoder over a complete inner iteration, i.e., across $\mathcal{I}$ iterations. Let $\bar{d}_v$ and $\bar{d}_c$ denote the average degree of the VNs and CNs for the component code, respectively. The number of operations is summarized as follows.

- 1) Computing input LLR requires N multiplications and N real additions.
- 2) The VNs updating requires $N\mathcal{I}(\bar{d}_v - 1)$ real additions. Specifically, for each VN, computing the message on one edge requires $\bar{d}_v - 1$ real additions. After computing the sum of all $\bar{d}_v$ incoming messages and input LLR with one additional real addition, the messages on the remaining $\bar{d}_v - 1$ edges can be efficiently obtained by subtraction, requiring another $\bar{d}_v - 1$ real additions.
- 3) The CNs updating requires $2M\mathcal{I}(\bar{d}_c - 1)$ multiplications and $2M\mathcal{I}\bar{d}_c$ table look-ups.
- 4) Computing the extrinsic information requires N real additions.

Let J_i denote the average number of outer iterations for $\mathbf{B}_i$. The computational complexity of SWD algorithm is summarized as follows.

- 1) All inner iterations require $N(2\mathcal{I}\bar{d}_v - \mathcal{I} + 2)T_{\text{all}}$ real additions, $(N + 2M\mathcal{I}(\bar{d}_c - 1))T_{\text{all}}$ multiplications and $2M\mathcal{I}\bar{d}_c T_{\text{all}}$ table look-ups, where $T_{\text{all}} = \sum_{i=1}^{L-w+2} m(w-1)J_i + \sum_{i=L-w+3}^L ((L-i+2)m + \min(i+w-L-3, \Upsilon-1)\min(m, M))J_i$.
- 2) Computing $\mathbf{S}_1$ and $\mathbf{S}_2$ require $2mMNJ_{\text{all}}$ binary additions and $m(2N-m)(J_{\text{all}} - J_1 - J_L) + 2m(N-m)J_1 + m(N+M)J_L$ comparisons, where $J_{\text{all}} = \sum_{i=1}^L J_i$.
- 3) Determining whether $\mathbf{S}_1 = \mathbf{S}_2 = \mathbf{0}_{M \times m}$ is satisfied requires $2mMJ_{\text{all}}$ comparisons.

Assume that the BP decoders are used as the cascaded decoders. Let J_0 denote the average number of outer iterations used in pre-decoding. T_e is the total average number of reliability enhancement, I_c is the maximum number of iterations per cascaded decoding, $\bar{I}_c$ is the average number of iterations per cascaded decoding, T_c is the average number of times the cascaded decoder, which is used during the PS-SWD process, T_p is the average number of modified input bits before each cascaded decoding. Compared to the SWD algorithm, the

additional computational complexity of C-SWD algorithm is summarized as follows.

- 1) Pre-decoding requires $2mMNJ_0$ binary additions, $mN(2\mathcal{I}\bar{d}_v - \mathcal{I} + 2)(w-2)J_0$ real additions, $m(N + 2M\mathcal{I}(\bar{d}_c - 1))(w-2)J_0$ multiplications, $2mM\mathcal{I}\bar{d}_c(w-2)J_0$ table look-ups and $2m(N + M - m)J_0$ comparisons.
- 2) Reliability enhancement requires $2mT_e$ multiplications.
- 3) Cascaded decoders require $MNT_c\bar{I}_c$ binary additions, $N(2\bar{d}_v - 1)T_c\bar{I}_c$ real additions, $2M(\bar{d}_c - 1)T_c\bar{I}_c + T_cT_p$ multiplications, $2M\bar{d}_cT_c\bar{I}_c$ table look-ups and $(M + N)T_c\bar{I}_c + J_{\text{all}}$ comparisons.

Assume that the OLDs are used as the cascaded decoders. For the i -th cascaded OLD decoding, let n_i denote the number of bits whose reliability has been enhanced by Eq.(34), E_i is the number of TEP used for re-encoding, R_i is the number of approximate likelihood, P_i is the number of integer partitions. Moreover, for the j -th integer partitions, let $w_{i,j}$ and $\bar{g}_{i,j}$ denote the Hamming weight and partition number, respectively. The computational complexity of i -th cascaded OLD is summarized as follows, where $E = \sum_{i=1}^{T_c} E_i$, $n = \sum_{i=1}^{T_c} n_i$ and $P = \sum_{i=1}^{T_c} P_i$.

- 1) The permutation Π_3 requires $(N - n_i)\log_2(N - n_i)$ comparisons.
- 2) Gaussian elimination requires $N\min(K, M)^2$ binary additions.
- 3) Multi-segment linear approximation requires $(3N - 3n_i + 2)(D - 1)$ real additions, $(N - n_i + 1)(D - 1)$ multiplications and $(N - n_i)D$ comparisons. Quantization requires $3D$ real additions, $4D$ multiplications and $4D - 1$ comparisons.
- 4) The TEPs generating criterion requires $7R_i + 5P_i + \sum_{j=1}^{P_i} (10\bar{g}_{i,j}w_{i,j} + 2\bar{w}_{i,j} - \bar{g}_{i,j})$ real additions, $7R_i + 7P_i + 2\sum_{j=1}^{P_i} \bar{g}_{i,j}w_{i,j}$ multiplications, 1 table look-up and $2R_i + P_i + \sum_{j=1}^{P_i} (2\bar{g}_{i,j}w_{i,j} - \bar{g}_{i,j})$ comparisons, where integer partitions operation requires $2P_i + 2\sum_{j=1}^{P_i} \bar{g}_{i,j}w_{i,j}$ multiplications, $\sum_{j=1}^{P_i} (8\bar{g}_{i,j}w_{i,j} + 2\bar{w}_{i,j} - \bar{g}_{i,j}) - P_i$ real additions and $P_i + \sum_{j=1}^{P_i} (2\bar{g}_{i,j}w_{i,j} - \bar{g}_{i,j})$ comparisons.
- 5) Re-encoding requires $K(M + 1)E_i$ binary additions. Determining the minimum WHD requires $(N - n_i - 1)E_i$ real additions and $E_i - 1$ comparisons.
- 6) Stopping criterion requires $M + E_i$ real additions, M multiplications, M table look-ups and E_i comparisons.

In the following, we analyze the computational complexity of three coding schemes for comparison. For PIC-LDPC codes [30], let $\mathcal{I}^p$ and $\mathcal{J}^p$ denote the maximum numbers of inner and outer iterations in the FF-FB decoding algorithm, respectively, and let J^p denote the average number of outer iterations. The computational complexity of the FF-FB algorithm is summarized as follows.

$$T_i = \begin{cases} \frac{(m^s + 1)(2w^s + 2 - m^s)}{2}, & i \in [1, L - w^s], \\ \frac{2(m^s + 1)(w^s + 1 - m^s) - (L - w^s + m^s - i)(L - m^s - w^s - i - 1)}{2}, & i \in [L - w^s + 1, L + m^s - w^s], \\ (m^s + 1)(L - i + 1) & i \in [L + m^s - w^s + 1, L]. \end{cases} \quad (37)$$

TABLE II
COMPUTATIONAL COMPLEXITY PER INFORMATION BIT

Schemes Operations	PS-LDPC codes (SWD algorithm)	PS-LDPC codes (C-SWD algorithm)	PIC-LDPC codes [30] (FF-FB algorithm)	SysConv-LDGM codes [31] (ISWD algorithm)	PSC-LDPC codes [32] (Full BP algorithm)
Binary additions	$2MNJ_{\text{all}}/((K-m)L)$	$MN(2mJ'_{\text{all}} + T_c\bar{I}_c)/((K-m)mL)$	$MNLJ^p/(L(K-m)-m)$	0	$(ZL(1+\alpha w_{\text{SC}})M_p N_p \bar{I}^b)/(LN_p - (L+\alpha w_{\text{SC}})M_p)$
Real additions	$N(2\bar{d}_v - \mathcal{I} + 2)T_{\text{all}}/((K-m)mL)$	$N((2\bar{d}_v - 1)T_c\bar{I}_c + (2\bar{d}_v - \mathcal{I} + 2)T'_{\text{all}})/((K-m)mL)$	$((2N\bar{I}^p\bar{d}_v + 4m - N\bar{I}^p)LJ^p)/(L(K-m)-m)$	$(J^s + 2\sum_i J_i^s(2T_i\bar{d}_o - V_i))/L$	$LN_p(2\bar{d}_v - 1)\bar{I}^b/(LN_p - (L+\alpha w_{\text{SC}})M_p)$
Multiplications	$(N + 2MT(\bar{d}_c - 1))T_{\text{all}}/((K-m)mL)$	$2M(\bar{d}_c - 1)\bar{I}_c + (N + 2MT(\bar{d}_c - 1))T'_{\text{all}}/((K-m)mL)$	$2MT^p(\bar{d}_c - 1)LJ^p/(L(K-m)-m)$	$(M\bar{d}_p(m^s(L-1) + 4\sum_i J_i^s T_i))/ (KL)$	$2(L + \alpha w_{\text{SC}})M_p(\bar{d}_t - 1)\bar{I}^b/(LN_p - (L + \alpha w_{\text{SC}})M_p)$
Comparisons	$((2N + 2M - m)J_{\text{all}} - (K-m)J_L - mJ_1)/((K-m)L)$	$((M+N)T_c\bar{I}_c + J_{\text{all}} + ((2N + 2M - m)J'_{\text{all}} - (K-m)J_L - m(J_0 + J_1)))/((K-m)mL)$	$((N + M - m)L - m)J^p/(L(K-m)-m)$	$1 + J^s/KL$	$((L + \alpha w_{\text{SC}})M_p + LN_p)\bar{I}^b/(LN_p - (L + \alpha w_{\text{SC}})M_p)$
Table look-ups	$2MT\bar{d}_c T_{\text{all}}/((K-m)mL)$	$2M\bar{d}_c(TT'_{\text{all}} + T_c\bar{I}_c)/((K-m)mL)$	$2MT^p\bar{d}_c LJ^p/(L(K-m)-m)$	$(4M\sum_i J_i^s(T_i\bar{d}_p + U_i) + KJ^s + Mm^s(L-1)(\bar{d}_p + 2))/ (KL)$	$2(L + \alpha w_{\text{SC}})M_p\bar{d}_t\bar{I}^b/(LN_p - (L + \alpha w_{\text{SC}})M_p)$
Total operations	$((2N + 2M + 2MN - m)mJ_{\text{all}} - m^2J_1 - (K-m)mJ_L + (3N + N\mathcal{I}(2\bar{d}_v - 1) + 2MT(2\bar{d}_c - 1))T_{\text{all}})/((K-m)mL)$	$((2MN + 2N + 2M - m)mJ'_{\text{all}} + (MN - M + 2N\bar{d}_v + 4M\bar{d}_c)T_c\bar{I}_c + T'_{\text{all}}(N(2\bar{d}_v - \mathcal{I} + 3) + 2MT(2\bar{d}_c - 1)) + 2mT_e + J_{\text{all}} - (K-m)mJ_L + T_cT_p - m^2(J_0 + J_1))/((K-m)mL)$	$((N + M + MN + 3m + N\bar{I}^p(2\bar{d}_v - 1) + 2MT^p(2\bar{d}_c - 1))LJ^p - mJ^p)/(L(K-m)-m)$	$1 + ((2K + 1)J^s + 2Mm^s(L-1)(\bar{d}_p + 1) + 2\sum_i J_i^s(2KT_i\bar{d}_o + 4MT_i\bar{d}_p - KV_i + 2MU_i))/ (KL)$	$(2LN_p\bar{d}_v + (L + \alpha w_{\text{SC}})(4M_p\bar{d}_t - M_p) + ZL(1 + \alpha w_{\text{SC}})M_p N_p)\bar{I}^b/(LN_p - (L + \alpha w_{\text{SC}})M_p)$

- 1) The FF decoding and FB decoding require $(2N\bar{I}^p\bar{d}_v - N\bar{I}^p + 4m)LJ^p$ real additions, $2MT^p(\bar{d}_c - 1)LJ^p$ multiplications and $2MT^p\bar{d}_c LJ^p$ table look-ups.
- 2) The FF and FB checks require $MNLJ^p$ binary additions and $((N + M - m)L - m)J^p$ comparisons.

For SysConv-LDGM codes [31], the systematic LDGM codes serve as the component code with $\bar{d}_v = \frac{K(M\rho^s+1)}{N}$ and $\bar{d}_c = \frac{K(M\rho^s+1)}{M}$, where ρ^s denotes the success probability of Bernoulli distribution. Let m^s denote the encoding memory, J^s is the maximum number of iterations in ISWD algorithm, w^s is the window size, and J_i^s is the average number of iterations for the i -th codeword. The computational complexity of ISWD algorithm is summarized as follows.

- 1) The forward and backward recursions require $4M\bar{d}_p \sum_{i=1}^L J_i^s T_i$ multiplications, $2K \sum_{i=1}^L J_i^s(2T_i\bar{d}_o - V_i)$ real additions and $4M \sum_{i=1}^L J_i^s(T_i\bar{d}_p + U_i)$ table look-ups, where $\bar{d}_o = \frac{N\bar{d}_v - M}{K(m^s+1)}$, $\bar{d}_p = \frac{\bar{d}_c - 1}{m^s+1}$, $V_i = \min(w^s+1, L-i+1)$, $U_i = \min(w^s+1, L+m^s-i+1)$, T_i is given by Eq. (37), shown at the bottom of the previous page.
- 2) Decision requires KL comparisons.
- 3) The entropy-based stopping criterion [45] requires KJ^s real additions, KJ^s table look-ups and J^s comparisons, where $J^s = \sum_{i=1}^L J_i^s$.
- 4) Cancellation requires $Mm^s(L-1)\bar{d}_p$ multiplications and $Mm^s(L-1)(\bar{d}_p + 2)$ table look-ups.

For PSC-LDPC codes [32], let $M_p \times N_p$ denote the size of base matrix, w_{SC} is the coupling width, and α is the proportion of SC-LDPC protograph retained. Then, the size of parity check matrix $\mathbf{H}^{\text{PSC}}$ is $Z(L + \alpha w_{\text{SC}})M_p \times ZLN_p$. By analogy with the previous analysis, we have $N = ZN_p$, $K = Z(N_p - M_p)$, where $\bar{d}_v$ and $\bar{d}_c$ are the average degrees of VNs and CNs for the underlying block LDPC codes, respectively. Thus, the average degrees of VNs and CNs for PSC-LDPC codes are $\bar{d}_v$ and $\bar{d}_t = \frac{L\bar{d}_c}{L + \alpha w_{\text{SC}}}$, respectively.

Let I^b and $\bar{I}^b$ denote the maximum and average number of iterations for full BP algorithm, respectively. Considering the structure of $\mathbf{H}^{\text{PSC}}$, the computational complexity of computing the syndrome can be reduced from $Z^2L(L + \alpha w_{\text{SC}})M_p N_p \bar{I}^b$ to $Z^2L(1 + \alpha w_{\text{SC}})M_p N_p \bar{I}^b$.

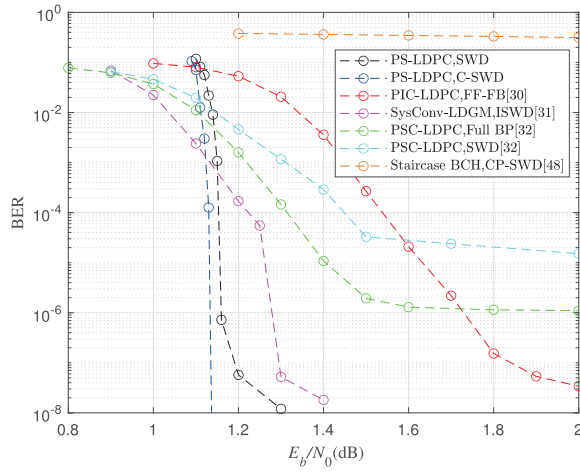
The total number of operations of above schemes are summarized in Table II, where $T'_{\text{all}} = T_{\text{all}} + m(w-2)J_0$ and $J'_{\text{all}} = J_{\text{all}} + J_0$.

V. SIMULATION AND DISCUSSION

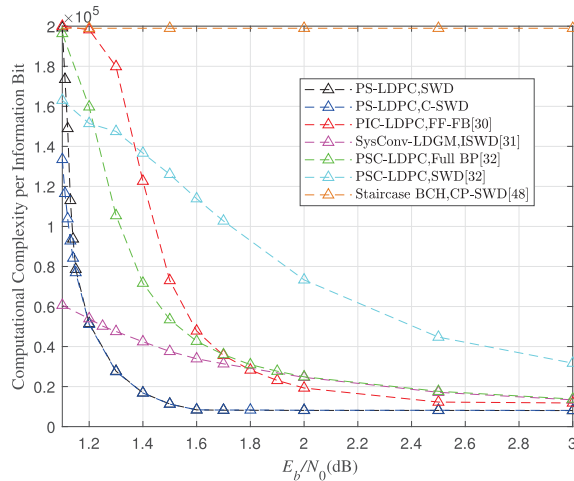
In this section, we conduct simulations to evaluate the decoding performance and complexity of the proposed PS-LDPC and C-SWD schemes. For the component codes of PS-LDPC codes, $N = 1056,960$ are constructed from the IEEE 802.16e standard, $N = 384$ is constructed from IEEE 802.22 standard, and $N = 256$ is obtained from CCSDS. We show the decoding complexity by comparing the average decoding complexity in terms of the number of operations per information bit [46].

TABLE III
SIMULATION CODING PARAMETERS OF DIFFERENT SCHEMES

Schemes	Parameters	N	K	R_d	L	R	$\mathbf{H}$
PS-LDPC codes		1056	704	$R_d = 0.5$ ($m = 352$) based on Eq. (2)	80	$R = 0.4969$ ($\Upsilon = 1$) based on Eq. (1)	IEEE 802.16e, 2/3A
PIC-LDPC codes [30]		1056	704	$R_d = \frac{K-m}{N-m}$ $= 0.5$ ($m = 352$)	81	$R = \frac{L(K-m)-m}{L(N-m)-m}$ $= 0.4969$	IEEE 802.16e, 2/3A
SysConv-LDGM codes [31]		1056	528	$R_d = \frac{K}{N} = 0.5$	480	$R = \frac{KL}{NL + m^s(N-K)}$ $= 0.4969$ ($m^s = 6$)	$\rho^s = 0.05$ [31]
PSC-LDPC codes [32]		1056	528	$R_d = \frac{K}{N} = 0.5$	81	$R = \frac{L(N_p - M_p) - \alpha w_{SC} M_p}{LN_p}$ $= 0.4969$ ($\alpha = 0.5$)	$\mathcal{E}_B(3, 6, L)$ [32], $Z = 528$



(a) BER.



(b) Complexity.

Fig. 9. BER performance and complexity comparisons.

The BER performance and complexity comparisons between the proposed PS-LDPC coding scheme and related schemes are illustrated in Fig. 9. The encoding parameters are listed in Table III to ensure that all schemes have the same codeword length N , design rate R_d , and actual code

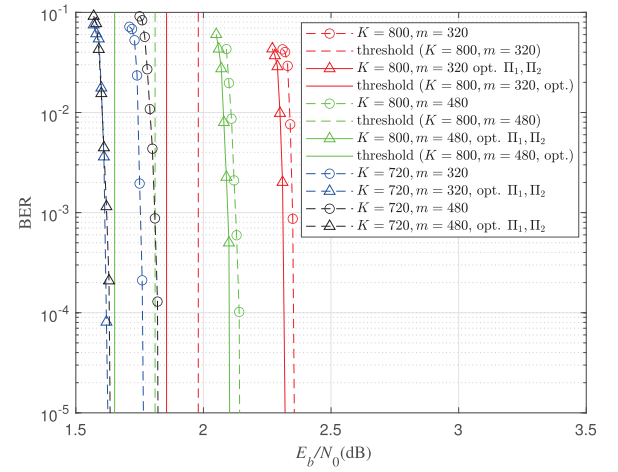
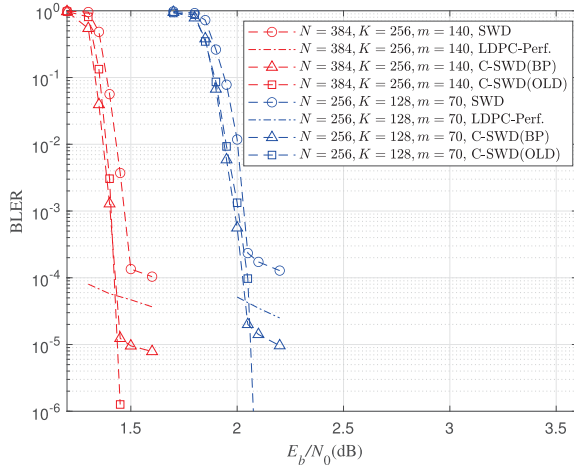
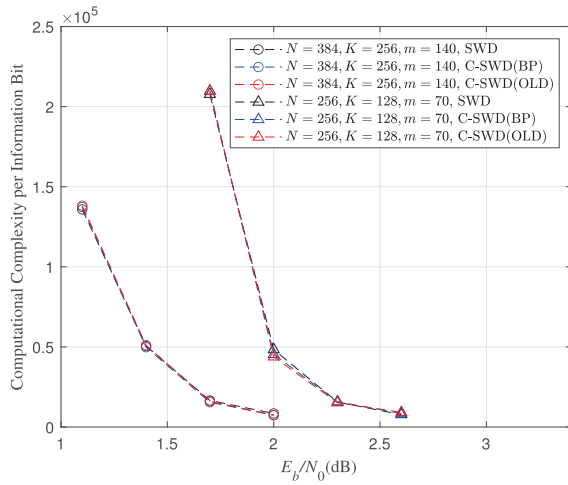


Fig. 10. BER performance comparison of C-SWD algorithm under different coupling patterns, where $N = 960$, $L = 40$, $\Upsilon = 1$, $w = 5$, $\mathcal{I} = 15$, $\mathcal{J} = 15$, $\varepsilon = 0.1$, $T_{\text{flag}} = 100$, $\kappa = 20$ and $I_c = 200$.

rate R . The decoding parameters are configured such that the maximum computational complexity per information bit for all schemes is approximately 2×10^5 . Specifically, for the SWD algorithm of PS-LDPC codes, set $\mathcal{I} = 18$, $\mathcal{J} = 25$, $w = 7$, and $\varepsilon = 0.1$. For the proposed C-SWD algorithm, set $\mathcal{I} = 18$, $\mathcal{J} = 17$, $w = 7$, $\varepsilon = 0.1$, $T_{\text{flag}} = 100$, $\kappa = 20$, and $I_c = 200$. For the FF-FB algorithm, set $\mathcal{I}^p = 192$ and $\mathcal{J}^p = 17$. For the ISWD algorithm, set $\mathcal{J}^s = 63$ and $w^s = 12$. For the full BP algorithm, set $I^b = 123$. For the SWD algorithm of PSC-LDPC codes, set the window size to 7 and the maximum number of iterations to 35. As shown in Fig. 9(a), under the same maximum decoding complexity, the PS-LDPC coding scheme achieves the BER $< 10^{-4}$ with the lowest E_b/N_0 , while the proposed C-SWD algorithm effectively mitigates the error floor. In addition, we evaluate the performance of staircase BCH code using the Chase-Pyndiah based SWD (CP-SWD) algorithm [47], with $N = 1023$, $m = 333$, $K = 678$ and primitive polynomial $x^{10} + x^3 + 1$. It can be observed that the staircase BCH code perform poorly. This is because, within the simulated SNR range, the number of errors far exceeds the error correction capability of the component BCH code. Fig. 9(b) further demonstrates that at higher E_b/N_0 , the PS-LDPC scheme exhibits the lowest decoding complexity, and at lower E_b/N_0 ,



(a) BLER.



(b) Complexity.

Fig. 11. BLER performance and complexity comparison between the SWD algorithm and the C-SWD algorithm with different cascaded decoders, where $L = 80$, $\Upsilon = 1$, $w = 7$, $\mathcal{I} = 10$, $\mathcal{J} = 30$, $\varepsilon = 0.1$, $T_{\text{flag}} = 20$, $\kappa = 10$, $I_c = 200$, $D = 2$ and $T'_{\text{flag}} = 10$.

the proposed C-SWD algorithm achieves lower complexity compared to the conventional SWD algorithm. It is worth noting that considering the location of the waterfall region, our PS-LDPC codes may gradually lose their advantage for larger N .

Then, we simulate the performance of C-SWD algorithm under the proposed optimal coupling pattern as shown in Fig. 10, and compare with the conventional encoding process. We can observe that the BER performance varies across different component codes. In particular, for the case of $K = 720$ and $m = 320$, the proposed OCP encoding algorithm achieves an improvement of approximately 0.15 dB. Although, due to finite-length scaling, the thresholds are further away from the BER performance compared to Fig. 6, it still demonstrates that DE is effective for determining optimal coupling patterns under short block-lengths.

Moreover, we compare the BLER performance and complexity between the SWD and C-SWD algorithms with different cascaded decoders in Fig. 11. Fig. 11(a) illustrates that while the cascaded OLD underperforms the cascaded BP decoder at low E_b/N_0 , it offers a significantly reduced

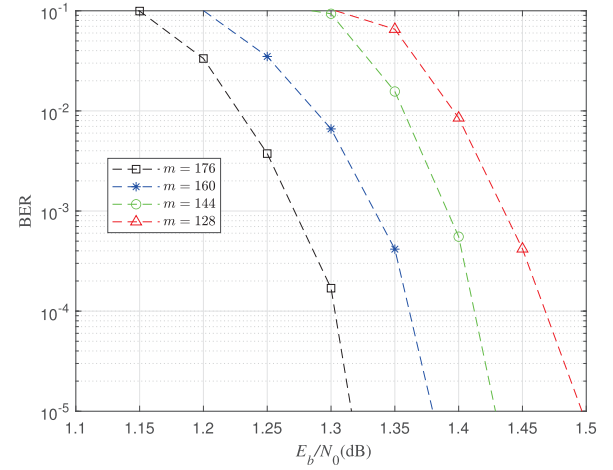


Fig. 12. BER performance of C-SWD algorithm under different m for the optimal coupling pattern, where $N = 384$, $K = 256$, $L = 40$, $\Upsilon = 1$, $w = 7$, $\mathcal{I} = 10$, $\mathcal{J} = 30$, $T_{\text{flag}} = 20$, $\kappa = 10$, $D = 2$ and $T'_{\text{flag}} = 10$.

error floor at high E_b/N_0 . Although the OLD has a relatively high complexity [48], Fig. 11(b) shows that cascaded OLD does not significantly increase the overall complexity of the C-SWD algorithm. This is primarily because the OLD is activated with a very low probability. For instance, the OLD is activated only about once per entire decoding process for $N = 384$, $K = 256$, $m = 140$ at $E_b/N_0 = 1.7$ dB. Additionally, for $m = 70$ and $E_b/N_0 = 2$ dB, the C-SWD algorithm achieves slightly lower decoding complexity, which can be attributed to the reduced average number of outer iterations.

Finally, Fig. 12 shows the decoding performance of C-SWD with cascaded OLD under different m for the optimal coupling pattern, where $\varepsilon = 0.2$ for $m = 176$ and $m = 160$, $\varepsilon = 0.1$ for $m = 144$ and $m = 128$. We can observe that by varying m , the code rate can be adjusted to accommodate different inter-satellite channel conditions. Meanwhile, a larger m yields better performance but also imposes higher parallel processing demands, which presents a design trade-off. Based on our simulation results of Fig. 10 and Fig. 12, we have identified two key characteristics of a well-performing coupling pattern for PS-LDPC codes. 1) The first group of VNs tend to exhibit the highest average node degree among the three groups. This is because these nodes generally possess more reliable a priori information, and VNs of higher degree exert greater influence during iterative message passing. However, the optimal coupling pattern does not simply maximize the degree of the first group of VNs. We observe that allocating some high degree nodes to the third group of VNs leads to further performance gains. 2) CNs tend to minimize the average number of edges connected from the second group of VNs. The information from uncoupled nodes is typically the least reliable. When multiple unreliable VNs connect to the same CN, this structure increases the likelihood that the parity-check equation may still be satisfied even if these nodes are in error, making the errors difficult to correct.

VI. CONCLUSION

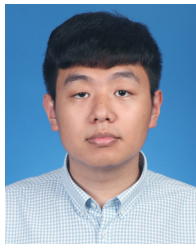
In this paper, we have proposed the rate-compatible PS-LDPC codes as a promising coding scheme for high speed inter-satellite communications. We firstly have characterized

the PS-LDPC code ensemble using three edge-types, and performed DE analysis under the Gaussian approximation to obtain threshold results. Based on the analysis, we have proposed the OCP encoding algorithm to achieve an optimal coupling pattern by adjusting the edge types connected to VNs. Furthermore, we have designed the C-SWD algorithm, which improves decoding performance and reduces the error floor through pre-decoding, reliability enhancement, and cascading decoders. Compared with other schemes, simulation results have shown that under the same maximum computational complexity per information bit, the proposed PS-LDPC coding scheme achieves the lowest BER and decoding complexity at high E_b/N_0 . Meanwhile, the cascaded OLD can more effectively suppress the error floor, meeting the high reliability requirements of high speed inter-satellite communications. To further enhance the rate-compatible ability, the advanced modulations for PS-LDPC scheme is studied for the future work.

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